



Advanced hybrid ensemble learning models for atmospheric humidity prediction: an environmental monitoring solution in arid regions

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Abstract

Atmospheric humidity significantly influences air quality, thermal comfort, and public health outcomes, particularly in arid regions highly impacted by climate change. This study develops innovative environmental monitoring techniques through advanced prediction models for atmospheric humidity to support air quality management and health risk assessment in Kuwait City. Analysis of a 40-year dataset (1984–2023) comprising eleven atmospheric parameters revealed significant correlations between humidity variations and heat stress conditions. Multicollinearity assessment using Variance Inflation Factor, enabled exclusion of highly correlated variables (Tavg: VIF = 92.14; PS: VIF = 8.49), thereby enhancing model reliability. Four ensemble learning models— Random Forest, XGBoost, LightGBM, and GBR were applied to predict daily humidity levels. RF demonstrated superior performance ($R^2 = 0.99329$, RMSE = 1.31601) in capturing humidity dynamics. A novel environmental prediction framework comprising five HELMs was developed, with the optimal model (HELM-05) achieving enhanced accuracy ($R^2 = 0.99364$, RMSE = 1.28124). These technological improvements in humidity prediction accuracy enable better assessment of health risks associated with high humidity conditions, particularly during extreme heat events. The developed monitoring methodology provides an evidence-based tool for atmospheric monitoring and public health planning in arid regions, facilitating targeted interventions to reduce humidity-related health risks. This research contributes to enhanced understanding of environmental monitoring systems in arid environments and their implications for regional air quality management strategies.

Keywords Air quality management · Atmospheric modeling · Climate technology · Environmental health · Environmental monitoring · Hybrid ensemble learning model

Abbreviations

EL Ensemble learning
HELMs Hybrid ensemble learning models
RF Random forest

XGBoost Extreme gradient boosting machine
LightGBM Light gradient-boosting machine
GBR Gradient boosting regression
VIF Variance inflation factor
LaRC Langley research centre
T_{avg} Average temperature

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$T_{\max}$	Maximum temperature
$T_{\min}$	Minimum temperature
SI	Solar irradiance
WS10M	Windspeed at 10 m
WS50M	Windspeed at 50 m
PS	Surface pressure
RH	Relative humidity
SH	Specific humidity
WBT	Wet bulb temperature
DFPT	Dew/frost point temperature
RMSE	Root mean square error
MAE	Mean absolute error
MAPE	Mean absolute percentage error
RAE	Relative absolute error

Introduction

Accurate weather prediction is crucial across agriculture, urban planning, and public health sectors, providing critical insights necessary for risk mitigation and informed decision-making. Among the various climatic variables, humidity levels are particularly important due to their direct and indirect health impacts (Gunawardhana et al. 2017a, b) and their pivotal role in shaping localized climate conditions (Farhat 2018). The prediction of humidity presents a complex challenge due to its dependence on complex, non-linear interactions within the climate system, such as temperature fluctuations, atmospheric pressure, and wind patterns (Lam et al. 2023; Wu et al. 2023). While traditional numerical weather prediction models form the cornerstone of meteorological forecasting, they often struggle to capture these intricate relationships, leading to inconsistent and sometimes unreliable results. As an alternative, ensemble learning (EL) techniques in machine learning have shown promise in improving prediction accuracy by combining the strengths of multiple models to address the inherent variability and non-linearity of climate systems (Singh et al. 2019; Dong et al. 2020; Salam and Islam 2020; Jose et al. 2022). These methods enable more robust and reliable forecasts, particularly for challenging variables like humidity (Tahsin et al. 2021; Lee et al. 2020). Furthermore, relative humidity plays a vital role in influencing the intensity of extreme weather events. Understanding these dynamics can inform the development of hybrid models that improve the accuracy of humidity level predictions in extreme climates.

High humidity levels reduce the body's ability to dissipate heat through sweat evaporation, leading to a phenomenon known as "reduced evaporative cooling" (Cramer et al. 2022). This physiological limitation intensifies the perceived temperature, often quantified by the heat index, thereby exacerbating the thermal sensation experienced by individuals and making the environment feel much hotter

than the actual air temperature (Sobolewski et al. 2021). Consequently, prolonged exposure to such conditions can lead to heat-related illnesses, including heat exhaustion, heatstroke, and cardiovascular stress (Cramer et al. 2022), posing critical health risks. In contrast, very low humidity can accelerate dehydration as the dry air absorbs moisture from the skin and respiratory system, compounding the risk of heat-related ailments (Wolkoff 2018). Populations, such as outdoor workers, the elderly, and individuals with pre-existing health conditions, are particularly vulnerable to these adverse effects of humidity variations (White-Newsome et al. 2012). Moreover, fluctuating humidity levels can create unpredictable patterns of heat stress, complicating efforts to mitigate these impacts effectively (Coffel et al. 2017; Javadinejad et al. 2020). Therefore, precise humidity forecasting is crucial for public health planning and developing early warning systems to protect populations in such varying climatic conditions.

In regions characterized by extreme climates, such as Kuwait, humidity plays a critical role in intensifying the effects of heat stress. Kuwait experiences a harsh desert climate characterized by prolonged periods of elevated temperatures and fluctuating humidity levels, particularly during the summer months (Alahmad et al. 2020). During summer, daytime temperatures often exceed 45 °C under intense solar radiation and clear skies, rendering the country one of the hottest places on Earth. Nighttime temperatures also remain unusually high. In contrast, rainfall is very sparse and irregular and occurring mainly in winter with an annual average of approximately 110 mm. This scarcity of rainfall, coupled with persistent high temperatures, often leads to excessive heat stress (Glaser et al. 2016; Alwadi and Alahmad 2024). Heat stress, in turn, poses substantial threats to public health, including dehydration, heatstroke, and other heat-related illnesses (Ncongwane et al. 2021; Sampath et al. 2023). The varying humidity levels can further exacerbate these risks, as elevated humidity impairs the body's ability to cool itself through evaporation, thereby increasing physiological strain and health hazards (Davis et al. 2016; Javadinejad et al. 2020). Beyond health impacts, humidity exerts influence across various factors, including hydrology, agriculture, and ecology by regulating key processes such as evaporation, transpiration, and moisture availability. Higher humidity levels can reduce evaporation rates from water bodies and soil while also affecting plant physiological responses, including leaf wetness duration, transpiration, and photosynthesis, which collectively impact nutrient uptake and overall plant vitality (Gunawardhana et al. 2017a, b). Despite its broad importance, humidity prediction has received comparatively less scholarly attention than other climatic variables like temperature and rainfall (Radhika and Shashi 2009; Ozbek et al. 2021; Kandasamy et al. 2025; Dong and Ding 2025). This gap underscores the urgent need for reliable forecasting



models to accurately predict humidity levels and support both climate adaptation strategies and public health interventions in extreme environments.

In the realm of climate prediction, traditional linear models such as linear regression and autoregressive approaches have historically provided foundational frameworks for forecasting weather-related variables due to their interpretability and straightforward implementation. However, these methods often struggle to capture the inherent nonlinearities and complex interdependencies present in meteorological data (Mendes-Moreira et al. 2012; Dietterich 2000). As a result, machine learning models have increasingly been explored for their ability to address these complexities, with ensemble learning (EL) models particularly Random Forest (RF), XGBoost, LightGBM, and Gradient Boosting Regression (GBR) emerging as leading methodologies (Mienye and Sun 2022; Sagi and Rokach 2018). Ensemble methods operate by aggregating the outputs of multiple learners, thereby reducing variance and bias, which mitigates the risks of overfitting that often compromise simpler models (Mienye and Sun 2022; Sagi and Rokach 2018). RF leverages decision tree ensembles to stabilize predictions against data outliers, leading to superior accuracy in predicting relative humidity (Mathur et al. 2024). Similarly, XGBoost's gradient boosting framework, noted for its speed and performance, effectively models nonlinear relationships in climatic data, outperforming traditional regressions in temperature and humidity forecasts (Ma et al. 2020; Mathur et al. 2024). LightGBM further advances these capabilities through histogram-based learning, excelling in large datasets and consistently reducing prediction errors (Zhang et al. 2024; Eom et al. 2020). Prior studies underscore that hybrid ensemble approaches, including stacking techniques, can integrate the strengths of RF, XGBoost, and LightGBM, achieving heightened accuracy in predicting not only humidity but also broader meteorological parameters such as precipitation and wind speed (Liang et al. 2021; Wang et al. 2023). Literature suggests that EL models represent state-of-the-art solutions in climate prediction, offering clear advantages over traditional linear frameworks in both theoretical grounding and practical application (Dietterich 2000; Sagi and Rokach 2018; Obisesan 2024).

The growing availability of climate data from advanced sensors, observational networks, and climate models has created new opportunities for improving environmental forecasts (Ladi et al. 2022). This data-rich environment, combined with computational advances, has paved the way for more sophisticated predictive techniques capable of capturing the intricate, nonlinear relationships within climate systems (Chen et al. 2023a). EL models stand out as particularly promising in this regard, offering a means to integrate multiple learning algorithms to achieve more robust and reliable predictions (Jose et al. 2022; Ahmed et al. 2020). Unlike single-model approaches, EL methods, through

techniques like bagging, boosting, and stacking are well suited to manage the inherent variability and uncertainty in climate data (Acharya et al. 2014; Crawford et al. 2019). Prior studies have demonstrated the utility of EL models in predicting weather variables such as temperature, air quality, and precipitation, highlighting their capacity to enhance forecast accuracy in dynamic climate systems (Nourani et al. 2019; Masood et al. 2023; Zhang et al. 2023; Lin et al. 2022; Yousaf et al. 2023; Liu and Yan 2024; Mantilla and Cárdenas 2024). However, despite these advances, humidity prediction has received less scholarly focus, particularly in regions like Kuwait where its interplay with temperature intensifies thermal stress (Slingo and Palmer 2011; Najafi et al. 2021). Addressing this gap is essential for improving operational climate services and public health preparedness in heat-vulnerable environments.

The main goal of this research is to improve the predictive capabilities of humidity forecasting systems in Kuwait City using the daily climate data from 1984 to 2023. This involves the development of innovative Hybrid Ensemble Learning Models (HELMs) for predicting humidity levels in the region. The study begins by evaluating the performance of several models, including Random Forest (RF), Light Gradient Boosting Machine (LightGBM), Gradient Boosting Regressor (GBR), and Extreme Gradient Boosting Machine (XGBoost). While most existing studies rely on standalone models or simple ensemble methods, the novelty of this paper lies in the development of optimized HELMs. These HELMs integrate multiple EL models within a unique stacking framework to improve humidity prediction accuracy in arid environments. Furthermore, this study integrated humidity forecasting with heat stress indicators, such as wet bulb temperature and dew/frost point temperature, to provide more meaningful applications for public health, urban planning, and climate adaptation strategies. The proposed models are designed for practical implementation, with potential integration into early warning systems, urban climate planning, and public health advisories in Kuwait.

Materials and methods

Study area

Kuwait City, the capital of Kuwait, is located between the coordinates of 29.3759° N latitude and 47.9774° E longitude, on the northeastern coast of the Arabian Peninsula, along the Persian Gulf (Fig. 1). The city plays a pivotal role in the nation's development, particularly in the oil industry and global trade, as the cultural, political, as well as economic center of Kuwait. It serves as the core of the Kuwait Metropolitan Area, which encompasses several neighboring towns and districts, contributing to its expansive urban



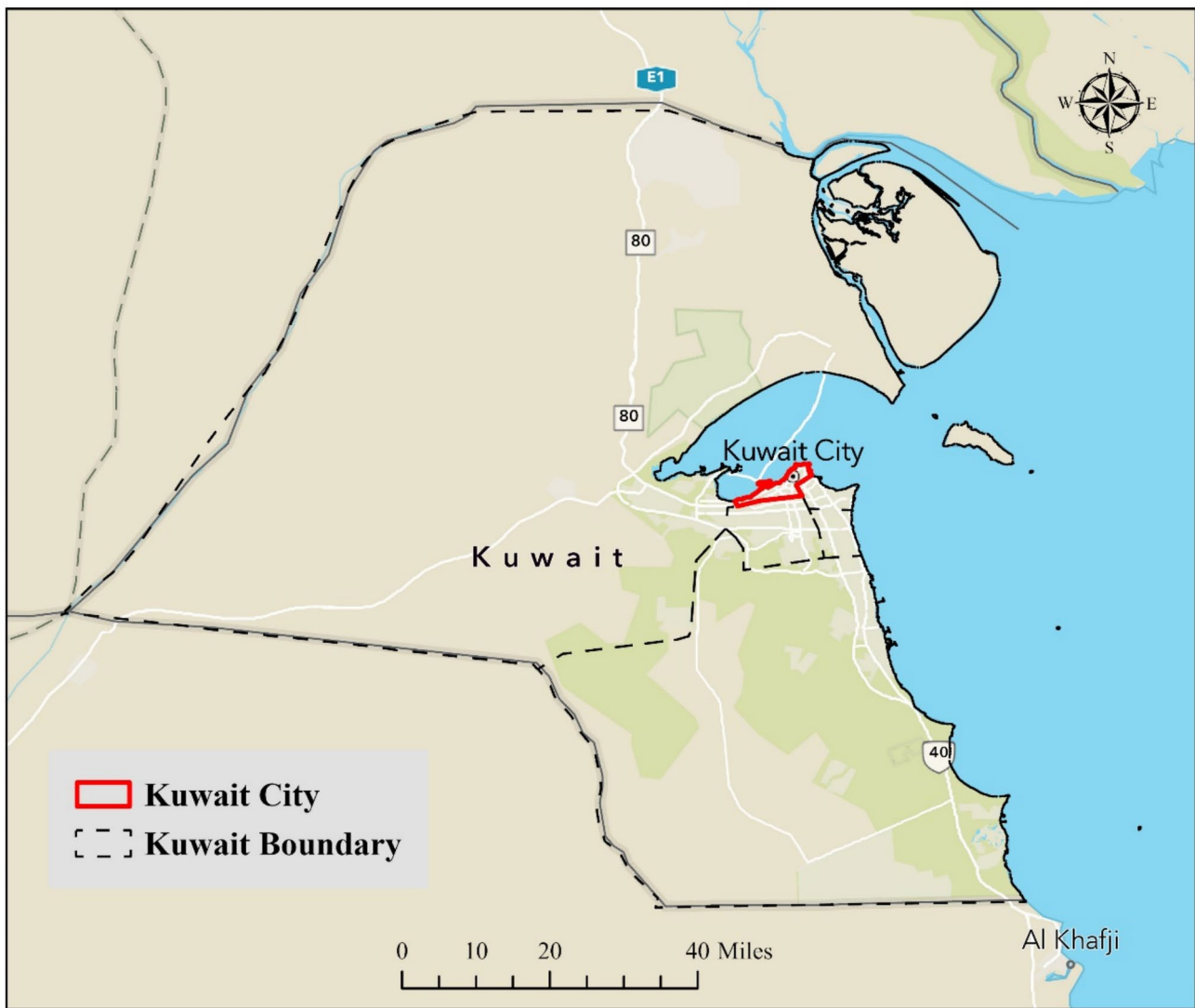


Fig. 1 Study area map of Kuwait City

sprawl. Over recent decades, Kuwait City has experienced rapid urbanization, now housing a population that exceeds 3 million people, a figure that continues to rise as a result of both migration and natural growth (AlDousari et al. 2023; Al-Dalamah and Al-Hurban 2019).

Geographically, the city is characterized by flat terrain extending into vast desert landscapes, salt flats, and arid regions that dominate the surrounding areas. The coastal location along the Persian Gulf is a defining feature of Kuwait City, which shapes its climatic conditions and influences urban development. Kuwait City experiences a desert climate marked by prolonged, extremely hot summers and mild winters. Summer temperatures frequently exceed 45 °C, particularly in July and August, with some days surpassing 50 °C during intense heatwaves, rendering the Kuwait city one of the hottest inhabited areas globally.

In contrast, winters are relatively mild, with temperatures ranging from 8 to 20 °C, offering temporary relief from the intense summer heat. Precipitation is sparse with an annual average of around 110 mm, mostly concentrated in the winter months. These arid climatic conditions, combined with the frequent occurrence of heatwaves, present substantial challenges to both the urban infrastructure and the health of the population, emphasizing the need for climate adaptation measures to address the extreme temperature fluctuations (Kamal et al. 2024; Alkandari 2024; Alahmad et al. 2023). The city's urban growth, coupled with its exposure to harsh climatic conditions, necessitates sustainable planning and development strategies that consider the ongoing impacts of climate change.



Data collection

This study used daily weather observations sourced from the NASA Langley Research Centre (LaRC), spanning the years 1984 to 2023. The dataset comprised key climatic variables, including average temperature (Tavg), maximum temperature (Tmax), minimum temperature (Tmin), solar irradiance (SI), wind speeds measured at both 10 m (WS10M) and 50 m (WS50M), wet bulb temperature (WBT), dew/frost point temperature (DFPT), surface pressure (PS), rainfall, specific humidity (SH) and relative humidity (RH). To ensure data accuracy and consistency, the collected daily data was transformed into monthly aggregates. For most variables, monthly averages were calculated, and then again cumulative totals were used for precipitation. These monthly values were then cross-indexed with local meteorological station data to validate the LaRC dataset's accuracy. Furthermore, this study conducted thorough data validation steps by examining both positive and negative values to verify data integrity and consistency. The statistical distribution was analyzed to confirm the applicability of the data for subsequent analyses, ensuring the reliability of the LaRC data as a solid foundation for this study.

Multicollinearity assessment and data preparation

In regression analysis, multicollinearity refers to a situation where two or more predictor variables exhibit a higher degree of correlation, suggesting that they convey overlapping information regarding the variability of the dependent variable. This interdependence among predictors can inflate the standard errors of the estimated regression coefficients, thereby complicating the interpretation of each predictor's unique impact on the outcome. High multicollinearity results in unreliable and unstable estimates of regression coefficients, as a result, impairing predictive accuracy (Sulaiman et al. 2021). To evaluate multicollinearity among the climatic variables, Variance Inflation Factor (VIF) and Tolerance were used, focusing particularly on the relationship between Tmax and the other climatic variables.

VIF quantifies the extent to which the variance of a regression coefficient is inflated due to multicollinearity, while Tolerance, its reciprocal, represents the proportion of variance in a predictor that is not explained by other predictors in the model. VIF and Tolerance were calculated using Eq. 1 and 2.

$$VIF_{\text{hum}} = \frac{1}{1 - R^2} \quad (1)$$

where R^2 is the coefficient of determination. The Tolerance for Tmax is calculated as:

$$\text{Tolerance}_{\text{hum}} = \frac{1}{VIF_{\text{hum}}} \quad (2)$$

A VIF value of 1 indicates no multicollinearity, suggesting that the predictor variables are not correlated. VIF values between 1 and 5 imply moderate multicollinearity, which may not significantly affect the model's accuracy but warrants monitoring. Values between 5 and 10 indicate high multicollinearity, suggesting that the variable might be redundant and not provide unique information. A VIF greater than 10 signifies very high multicollinearity, indicating that such predictors need careful evaluation before inclusion in the model. Similarly, tolerance values close to 1 suggest little to no multicollinearity, whereas values approaching 0 indicate severe multicollinearity. These metrics are essential for refining regression models by ensuring that each predictor uniquely contributes to the analysis and does not redundantly inflate the model's error (Miles 2005).

Gradient boosting regression

GBR is a boosting algorithm within aggregate learning techniques that construct models sequentially, with each new model aimed at correcting errors from previous iterations (Zhao et al. 2019). It is highly efficient in capturing complex, non-linear relationships within data, making it popular across various domains such as finance, health-care, and environmental modeling (Alonso et al. 2015). GBR integrates the predictions of multiple weak learners, typically decision trees, into a unified, strong predictive model. In contrast to single-pass regression models, GBR incrementally builds an ensemble by iteratively adding new models to address residual errors from previous models. This process, gradient boosting, carefully refines predictions by minimizing the model's loss function, which measures the discrepancy between actual and predicted values (Bentéjac et al. 2021; Nie et al. 2021).

The primary goal of GBR is to iteratively minimize the loss function $L(y, \hat{y})$, which denotes the error between the observed values y and predicted values $\hat{y}$. The prediction for an input x_i at iteration M is defined as Eq. 3:

$$\hat{y}_i = \sum_{m=1}^M f_m(x_i) \quad (3)$$

where $f_m(x_i)$ is the prediction from the m^{th} tree, and M denotes the total number of trees in the ensemble.



At each iteration m , GBR upgrades the model by adding a new tree trained to minimize the residuals $r_i^{(m)}$ which are the differences between the actual values y_i and the current model's predictions $\hat{y}_i^{(m-1)}$. These residuals are calculated using Eq. 4:

$$r_i^{(m)} = y_i - \hat{y}_i^{(m-1)} \quad (4)$$

The new tree $f_m(x_i)$ is trained to reduce the residuals by minimizing the loss function $L(y_i, f_m(x_i))$. The prediction is updated at each step in through Eq. 5.

$$\hat{y}_i^{(m)} = \hat{y}_i^{(m-1)} + \vartheta f_m(x_i) \quad (5)$$

where ϑ is the learning rate, that controls the contribution of each tree to the overall prediction and prevents overfitting by slowing the learning process.

GBR optimizes the objective function to enhance prediction accuracy and control overfitting, as denoted by Eq. 6:

$$Obj = \sum_{i=1}^n L(y, \hat{y}) + \sum_{m=1}^M \Omega(f_m) \quad (6)$$

Here, $\Omega(f_m)$ indicates the regularization term, which penalizes the complexity of the model to prevent overfitting by constraining the growth of the trees. The final GBR prediction $f(x)$ is the summation of the predictions from all M trees can be expressed by Eq. 7:

$$f(x) = \sum_{m=1}^M f_m(x). \quad (7)$$

Light gradient-boosting machine

The LightGBM is a potent and efficient EL algorithm, specifically designed for regression tasks and well-suited for handling large datasets with complex features. Particularly, it is particularly adept at processing time series data, making it valuable for applications such as climate modeling. LightGBM is renowned for its speed and accuracy, allowing it to handle extensive data efficiently (Taha et al. 2020). It constructs an ensemble of decision trees where each tree is sequentially trained to correct the errors of its predecessors. This iterative method is called boosting, in which subsequent trees minimize the residual errors from previous models. In contrast to traditional gradient boosting methods, LightGBM incorporates several key innovations that enhance its performance. Discretizing continuous features to reduce divides and using a leaf-wise growth approach to focus on the leaf with the best error reduction potential speeds training in LightGBM, enhancing efficiency and model complexity (Guo and Chang 2022). Hyperparameters are crucial in

determining LightGBM's performance. The learning rate controls the tree's contribution, the number of estimators balances the fit, the maximum depth limits complexity, the leaf's minimum data reduces overfitting, along the feature fraction incorporates randomness to improve generalization. To optimize the combination of these hyperparameters, methods such as Bayesian Optimization are often employed, which systematically searches for the best parameter values, improving model performance and efficiency (Fan et al. 2019). The final prediction for an input p in LightGBM is expressed as a weighted sum of the outputs from each weak learner $h_i(p)$, as in Eq. 8:

$$f(p) = \sum_{i=1}^N \omega_i \times h_i(p) \quad (8)$$

where N =the total number of weak learners (trees), and ω_i =the weight assigned to the i -th learner that indicates its contribution to the overall prediction. LightGBM's combination of boosting with advanced techniques like histogram-based learning and leaf-wise tree growth, as well as optimized hyperparameter tuning, makes it a fast and accurate model, particularly well-suited for high-dimensional and large-scale datasets.

Extreme gradient boosting machine

The XGBoost is a remarkably efficient and robust implementation of gradient boosting, known for its speed, scalability, and versatility in both regression and classification tasks. XGBoost's ability to handle large datasets with high efficiency, and its support for out-of-core computation, makes it a preferred algorithm among data scientists (Chen 2015). Its excellent performance is due to a combination of innovations, including regularization and optimization techniques which enhance predictive accuracy and reduce overfitting (Guo and Chang 2022; Alshboul et al. 2022; Carmona et al. 2019).

To comprehend XGBoost's mechanics, consider a dataset DS with n instances and m features, defined by Eq. 9.

$$DS = \{(x_i, y_i) : i = 1, \dots, n; x_i \in \mathbb{R}^m; y_i \in \mathbb{R}\} \quad (9)$$

In XGBoost, the final prediction $\hat{y}_i$ for an input x_i is computed by Eq. 10.

$$\hat{y}_i = \phi(x_i) = \sum_{k=1}^K f_k(x_i), f_k \in \mathcal{F} \quad (10)$$

where f_k represents the k -th tree in the ensemble and $\mathcal{F}$ is the space of possible trees. XGBoost attempts to find an optimal set of trees by minimizing a combination of the loss function and a regularization term. The objective function $L(\phi)$ incorporates both the training loss as well as the



regularization term to control model complexity which is expressed by Eq. 11.

$$L(\phi) = \sum_i l(y_i, \hat{y}_i) + \Omega(f_k) \quad (11)$$

Here, $l(y_i, \hat{y}_i)$ indicates the loss function that measures the discrepancy between predicted and actual values. The regularization term $\Omega(f_k)$ helps to lessen overfitting by penalizing complex models, which is calculated by Eq. 12.

$$\Omega(f_t) = \gamma T + 0.5 \lambda \omega^2 \quad (12)$$

where T is the number of leaves in the tree, ω denotes the weights associated with each leaf, and γ and λ are hyperparameters that control the strength of regularization. XGBoost implements a boosting technique where new trees are sequentially added to minimize the overall objective function. At each iteration t , a new function f_t is introduced to improve the model's accuracy by reducing the residuals from previous predictions by Eq. 13.

$$L^{(t)} = \sum_{i=1}^n l(y_i, \hat{y}_i^{(m-1)} + f_t(x_i)) + \Omega(f_t) \quad (13)$$

This iterative process gradually reduces errors by training each subsequent tree on the residuals of the previous model, ultimately refining the predictions. During training, the algorithm selects the optimal split for each node based on the split gain L_{split} , calculated by Eq. 14.

$$L_{split} = 0.5 \times \left[\frac{(\sum_{i \in I_L} g_i)^2}{\sum_{i \in I_L} h_i + \lambda} + \frac{(\sum_{i \in I_R} g_i)^2}{\sum_{i \in I_R} h_i + \lambda} - \frac{(\sum_{i \in I} g_i)^2}{\sum_{i \in I} h_i + \lambda} \right] - \gamma \quad (14)$$

Here, g_i and h_i indicates the first and second derivatives of the loss function with respect to the prediction, respectively. I_L and I_R represents the sets of data points in the left and right child nodes after the split. λ regulates the regularization applied to each split, then again γ penalizes the number of leaf nodes to prevent overfitting.

XGBoost is a combination of advanced techniques such as regularization, split optimization, and sequential training which enable it to deliver high predictive performance and computational efficiency. This makes it an ideal choice for complex and large-scale datasets.

Random forest

The RF algorithm is a recognizable flexible and powerful EL method that is mostly used for regression and classification. It extends the decision tree model by constructing a multitude of decision trees and their outputs are combined to boost predictive accuracy and manage overfitting issues

(Boateng et al. 2020). This ensemble approach is especially effective with high-dimensional data, as it can capture complex interdependencies between features. RF operates by constructing a series of decision trees during the training phase, where each tree contributes to the final prediction. In the case of regression, the RF's prediction is the average of the outputs from all trees. The RF trees are constructed using bootstrapped subsets of the original data, a process called bagging (Bootstrap Aggregating). By training each tree on a unique subset of the data, RF introduces randomness into the process which reduces overfitting and enhances the model's generalization capabilities (Alonso et al. 2015; Fan et al. 2022). Furthermore, each tree considers only a random subset of features at each split that ensures diversity among the trees. For regression tasks, RF averages the predictions from all individual decision trees, which is denoted as $T_m(x)$, to generate the final output $\hat{y}$. This average is calculated by Eq. 15:

$$\hat{y} = \frac{1}{M} \sum_{m=1}^M T_m(x) \quad (15)$$

where M is the total number of trees in the forest, and $T_m(x)$ is the prediction from the m -th tree. The predictions are averaged, and then RF minimizes variance and stabilizes the output, leading to more reliable predictions. This approach provides several key benefits like improved accuracy, overfitting control, and handling high-dimensional data (Ruiz-Alvarez et al. 2021; Smarra et al. 2018).

HELMS

Building on the performance evaluation of the initial EL models, this study developed five HELMs to further enhance prediction accuracy. These hybrid models strategically combine the strengths of individual ensemble models such as RF, XGBoost, LightGBM, and GBR. By leveraging model architecture and learning techniques, the hybrid models aim to minimize prediction errors and improve generalizability.

Accuracy assessment

The predictive performance of the models was assessed using five widely recognized statistical metrics: Coefficient of Determination (R^2), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Relative Absolute Error (RAE), and Mean Absolute Percentage Error (MAPE) using Eqs. 16–20 (González-Sopeña et al. 2021). These metrics offered a comprehensive assessment of accuracy, error magnitude, and reliability.



$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (16)$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n}} \quad (17)$$

$$MAE = \frac{\sum_{i=1}^n |y_i - \hat{y}_i|}{n} \quad (18)$$

$$RAE = \frac{\sum_{i=1}^n |y_i - \hat{y}_i|}{\sum_{i=1}^n |y_i - \bar{y}|} \quad (19)$$

$$MAPE = \frac{100\%}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \quad (20)$$

R^2 measured the proportion of variance in observed RH values explained by the model, with values closer to 1 indicating better fit. RMSE evaluated the average magnitude of prediction errors, assigning a higher weight to

larger errors, while MAE quantified the average absolute difference between observed and predicted values. RAE compared the model's performance to a naive baseline model, and MAPE expressed the average percentage difference between observed and predicted values, providing interpretable accuracy percentages.

Results and discussion

Variability of humidity levels

Figure 2 represents the recorded monthly variability of RH levels in Kuwait City during the study period. Outcomes revealed that RH exhibited significant seasonal variability over the period from 1984 to 2023, with monthly descriptive statistics revealing a pattern consistent with a desert climate. January and December experienced the highest average RH values at 63.56% and 61.34%, respectively, and also showed substantial daily variability (std of 11.33% and 11.16%), reflecting cooler and more humid winter conditions. In contrast, the summer months of June, July, and August recorded the lowest average RH values at 29.26%,

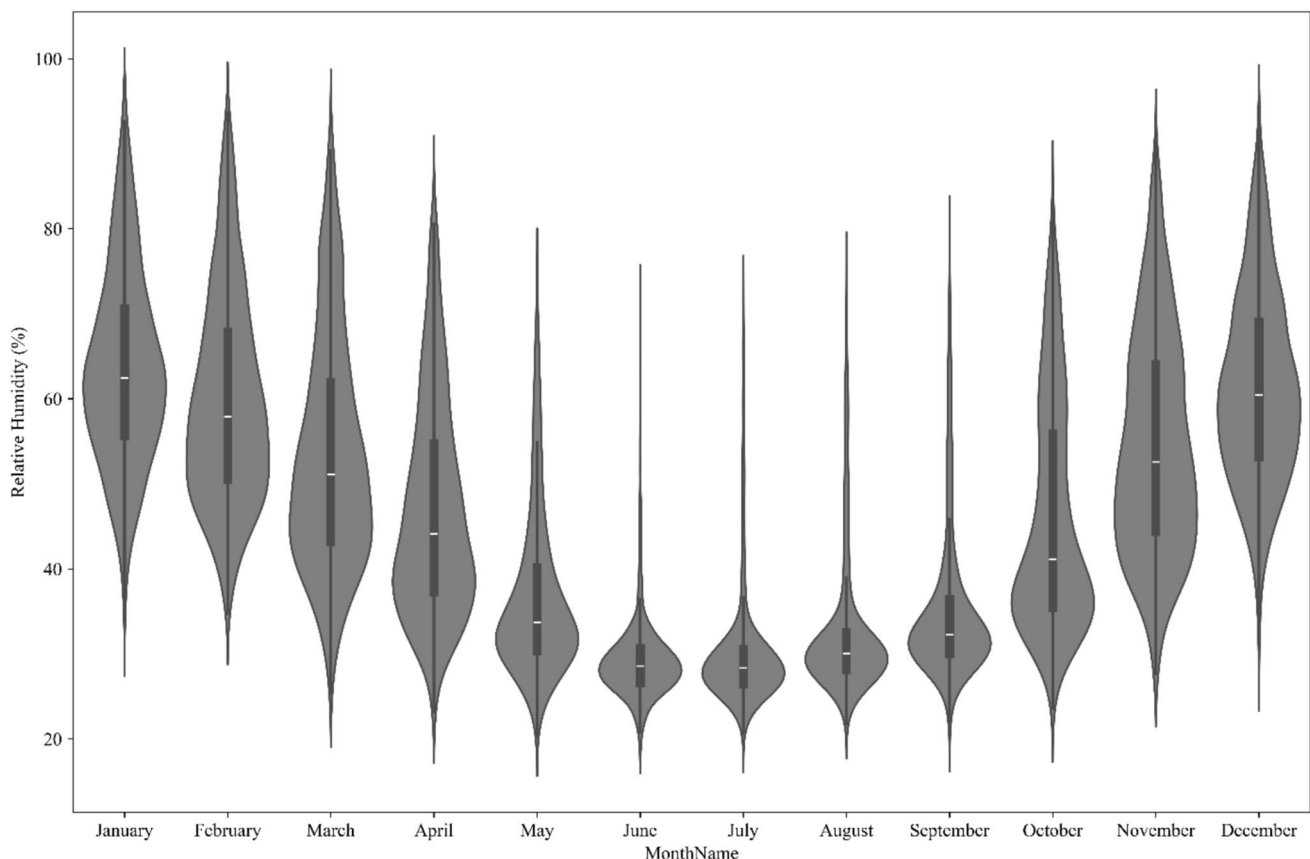


Fig. 2 Violin plot showing monthly variations in humidity levels in Kuwait City



30.05%, and 32.75%, respectively, with correspondingly lower variability (std of 5.02%, 7.19%, and 8.55%), indicating the extreme dryness typical of peak summer in a desert. Transition months like April and October presented intermediate average RH values of 46.81% and 45.73% with higher variabilities of 12.47% and 13.19%, signaling the shifting seasons with wider fluctuations in humidity levels. Moreover, November and March transitioned into cooler conditions with averages at 54.51% and 53.42% respectively, showcasing an increase in humidity as temperatures began to drop. These patterns highlighted how RH in Kuwait City varied significantly throughout the year, with the highest humidity and variability occurring during the cooler months and drier conditions with less fluctuation prevailing during the hot summer months.

Multicollinearity assessment of climate variables

The multicollinearity analysis revealed varying degrees of correlation among the climatic variables, as shown in Table 1. The majority of variables showed acceptable VIF values below 5.00 and Tolerance values above 0.2, indicating their suitability as independent predictors. SI showed minimal multicollinearity with a VIF of 1.23 and Tolerance of 0.81, followed by WS50M (VIF = 1.20, Tolerance = 0.83) and Rainfall (VIF = 1.11, Tolerance = 0.90). However, two variables exhibited concern levels of multicollinearity: Tavg with an exceptionally high VIF of 92.14 and very low Tolerance value of 0.01, and PS with a VIF of 8.49 and Tolerance of 0.12. Based on these findings, Tavg and PS were excluded from subsequent modeling to ensure robust predictions.

Performance analysis of individual EL models

The comparative analysis of the four ensemble learning models revealed distinct performance patterns, as detailed

Table 1 Outcomes of multicollinearity assessment

Climatic factor	Tolerance	VIF
SI	0.81	1.23
WS10M	0.28	3.57
WS50M	0.83	1.20
SH	0.44	2.27
Tmax	0.55	1.83
Tmin	0.57	1.76
Tavg	0.01	92.14
Rainfall	0.90	1.11
WBT	0.01	2.02
DFPT	0.03	31.80
PS	0.12	8.49

in Table 2. Outcomes show that the RF model exhibited highest precision ($R^2 = 0.99329$) in daily humidity level prediction, indicating that it can explain approximately 99.33% of the variance in humidity levels, which reflects its superior capability to fit the daily humidity data in Kuwait City. This is closely followed by the XGBoost and LightGBM models, which have R^2 values of 0.99248 and 0.99216, respectively, demonstrating similarly high levels of data fitting. GBR trails with an R^2 of 0.97957, showing it to be slightly less effective at modeling the variance compared to its counterparts. When comparing the models based on error metrics, RF again stands out, exhibiting the lowest error rates across all metrics with an RMSE of 1.31601, MAE of 0.88446, RAE of 0.06408, and MAPE of 1.90947. These figures highlight its robustness and accuracy in humidity prediction. In contrast, GBR records higher errors with an RMSE of 2.29598, MAE of 1.61574, RAE of 0.11706, and MAPE of 3.52462, indicating it has the largest prediction deviations among the models. Both XGBoost and LightGBM display intermediate performance, XGBoost slightly edges out LightGBM in minimizing errors, making it the second most reliable model following RF. These quantitative findings are visually corroborated in Fig. 3, where scatter plots reveal a tight clustering of predicted versus observed humidity values along the 1:1 reference line for RF, XGBoost, and LightGBM, confirming their strong fit. In contrast, the GBR model shows a more scattered distribution, indicating higher residual errors and less consistent prediction. Overall, the analysis demonstrates the robustness of RF model in modeling daily humidity level in arid environments.

Development and evaluation of hybrid ensemble models

Five hybrid ensemble models were constructed through strategic combinations of the base models, as outlined in Table 3. Table 4 summarizes the performance of five HELMs for predicting daily humidity levels in Kuwait City. Among them, HELM-05, which integrates all four base models (XGBoost, GBR, LightGBM, and RF), exhibited the highest predictive precision, achieving an R^2 of 0.99364, indicating its ability to explain about 99.36% of the variance in the humidity. Moreover, HELM-05

Table 2 Evaluation of the performance of EL models

Models	R^2	RMSE	MAE	RAE	MAPE
GBR	0.97957	2.29598	1.61574	0.11706	3.52462
LightGBM	0.99216	1.42202	1.01337	0.07342	2.22004
XGBoost	0.99248	1.39284	1.00101	0.07252	2.17605
RF	0.99329	1.31601	0.88446	0.06408	1.90947



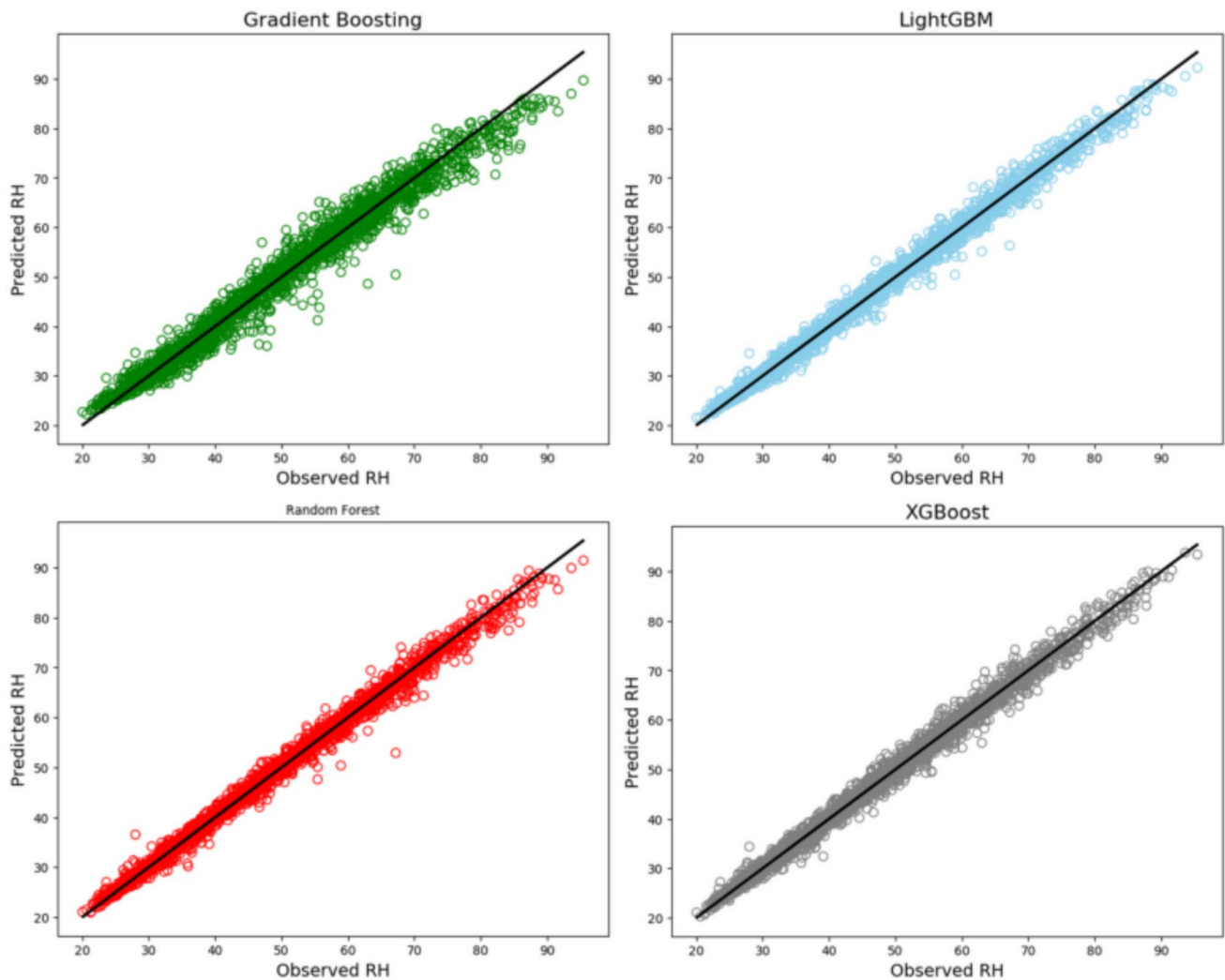


Fig. 3 Scatter plots of daily predicted vs observed relative humidity in Kuwait using EL models

Table 3 Overview of the combination of HELMs

Combination	Name
XGBoost + GBR + RF	HELM-01
XGBoost + LightGBM + RF	HELM-02
LightGBM + GBR + RF	HELM-03
XGBoost + GBR + LightGBM	HELM-04
XGBoost + GBR + LightGBM + RF	HELM-05

recorded the lowest error values across all performance metrics (RMSE of 1.28124, MAE of 0.88844, RAE of 0.06437, and MAPE of 1.9317) suggesting its superior accuracy and consistency in predicting daily humidity level in Kuwait City. In comparison, the models composed of the three base learners showed relatively poor performance. HELM-01 (XGBoost + GBR + RF) and HELM-02 (XGBoost + LightGBM + RF) have similar performance

Table 4 Evaluation of the performance of HELMs in predicting humidity levels in Kuwait City

Model Name	R^2	RMSE	MAE	RAE	MAPE
HELM-01	0.99334	1.31076	0.90837	0.06581	1.97096
HELM-02	0.99325	1.31946	0.90998	0.06593	1.97395
HELM-03	0.99303	1.34136	0.9311	0.06746	2.01994
HELM-04	0.99312	1.33237	0.94176	0.06823	2.05604
HELM-05	0.99364	1.28124	0.88844	0.06437	1.9317

metrics, with R^2 slightly over 0.993, indicating strong model fit, yet they showed marginally higher error metrics compared to HELM-05, suggesting slightly reduced precision. HELM-03 (LightGBM + GBR + RF) and HELM-04 (XGBoost + GBR + LightGBM) showed somewhat lower accuracy, as reflected in their elevated RMSE, MAE, and MAPE values (Table 4). These results underscore the



importance of integrating a broader set of diverse base models, as done in HELM-05, to achieve enhanced generalization and predictive reliability in complex climatic conditions.

The scatter plots for all five hybrid models, as presented in Fig. 4, shows strong alignment between predicted and observed humidity values. HELM-05 shows the tightest clustering along the 1:1 regression line, further confirming its superior performance, as indicated by the statistical performance metrics. The plots reveal minimal systematic bias across all models, with predictions distributed evenly around the isoline across the full range of humidity values. This visual analysis supports quantitative metrics and provides additional confidence in the models' predictive capabilities.

Discussion

Humidity variability profoundly influences human health, particularly in arid regions like Kuwait City, where extreme temperature fluctuations and significant humidity variations exacerbate heat stress and related illnesses (Wolkoff et al. 2021; Alahmad et al. 2023). Accurate prediction of humidity levels is thus critical for public health management, urban planning, and emergency response systems. This study employed EL models, namely Random Forest, XGBoost, LightGBM, and GBR to predict daily humidity variability in Kuwait City. Additionally, this study developed Hybrid Ensemble Learning Models (HELMs) to enhance predictive precision. The dataset comprised eleven meteorological variables, from which highly collinear variables (Tavg and PS) were excluded following established practices in environmental modeling (Sulaiman et al. 2021). This preprocessing step was crucial for model stability, as high multicollinearity can lead to unreliable predictions in complex environmental systems. The retained variables exhibited appropriate independence, providing a robust foundation for the ensemble learning approaches, aligning with findings from recent climate modeling studies (Lam et al. 2023; Wu et al. 2023).

The performance metrics indicated that all models achieved acceptable prediction accuracy, with RF exhibiting the highest precision ($R^2 = 0.99329$), followed by XGBoost ($R^2 = 0.99248$), LightGBM ($R^2 = 0.99216$), and GBR ($R^2 = 0.97957$). RF's superior performance is attributable to its ability to handle non-linear relationships and capture complex interactions between climate variables, corroborating findings by Boateng et al. (2020) and Fan et al. (2022). The relative success of LightGBM and XGBoost compared to GBR, reflects similar patterns observed in recent studies, where these advanced boosting algorithms demonstrated robust performance in handling environmental data (Chen

2015; Taha et al. 2020). The consistent performance across multiple evaluation metrics reinforces the reliability of these models for humidity prediction in arid climates. These findings align with previous research employing EL models for climatic predictions. For instance, Ma et al. (2020) utilized XGBoost to predict outdoor air temperature and humidity, achieving high accuracy and demonstrating the model's capability in handling complex environmental datasets. Similarly, Hussain et al. (2024) indicated that ensemble-based regression models, including Gradient Boosting frameworks, considerably improved predictions of daily temperature and rainfall patterns, underscoring the efficacy of machine learning in capturing nonlinear trends typical of climatic phenomena.

The development of HELM-05, which achieved an R^2 of 0.99364, represents a significant advancement in humidity prediction modeling. Unlike existing studies that rely solely on standalone models or simple ensemble techniques, HELM-05 integrates four base learners within a unified framework, leveraging heterogeneous learning behaviors to improve generalization and resilience against overfitting, a common challenge in arid climate datasets. This methodological advancement aligns with recent trends in hybrid modeling approaches, as demonstrated by Masood et al. (2023) and Zhang et al. (2023). The success of the hybrid approach can be attributed to the complementary strengths of different algorithms, improved handling of data variability, and enhanced robustness against individual model weaknesses. The systematic improvement in prediction accuracy achieved through hybrid modeling suggests that this approach could be valuable for other environmental prediction tasks in similar climatic contexts. Supporting literature further emphasizes the efficacy of hybrid models. For example, Hasan et al. (2024) presented a comparative analysis of several machine learning models, including recurrent neural networks (RNN) with long short-term memory (LSTM) and TensorFlow Decision Forest (TFDF), alongside a stacking strategy that integrates models like ElasticNet, GradientBoost, KRR, and Lasso. The results indicated that the stacking model significantly improved forecasting accuracy while addressing overfitting concerns. Additionally, Ren et al. (2022) elaborated on using deep neural networks, specifically LSTM and CNN, integrated through a stacking ensemble technique to enhance sand-dust storm forecasting, demonstrating enhanced predictive capabilities.

Despite the promising results, several challenges persist in predicting humidity and related climatic variables using machine learning and ensemble models. One issue is overfitting, where models become excessively complex and capture noise rather than underlying trends, particularly problematic in high-dimensional datasets with spurious correlations (Donat et al. 2024; Rader and Barnes 2023). Data sparsity, especially in regions with limited observational data, further



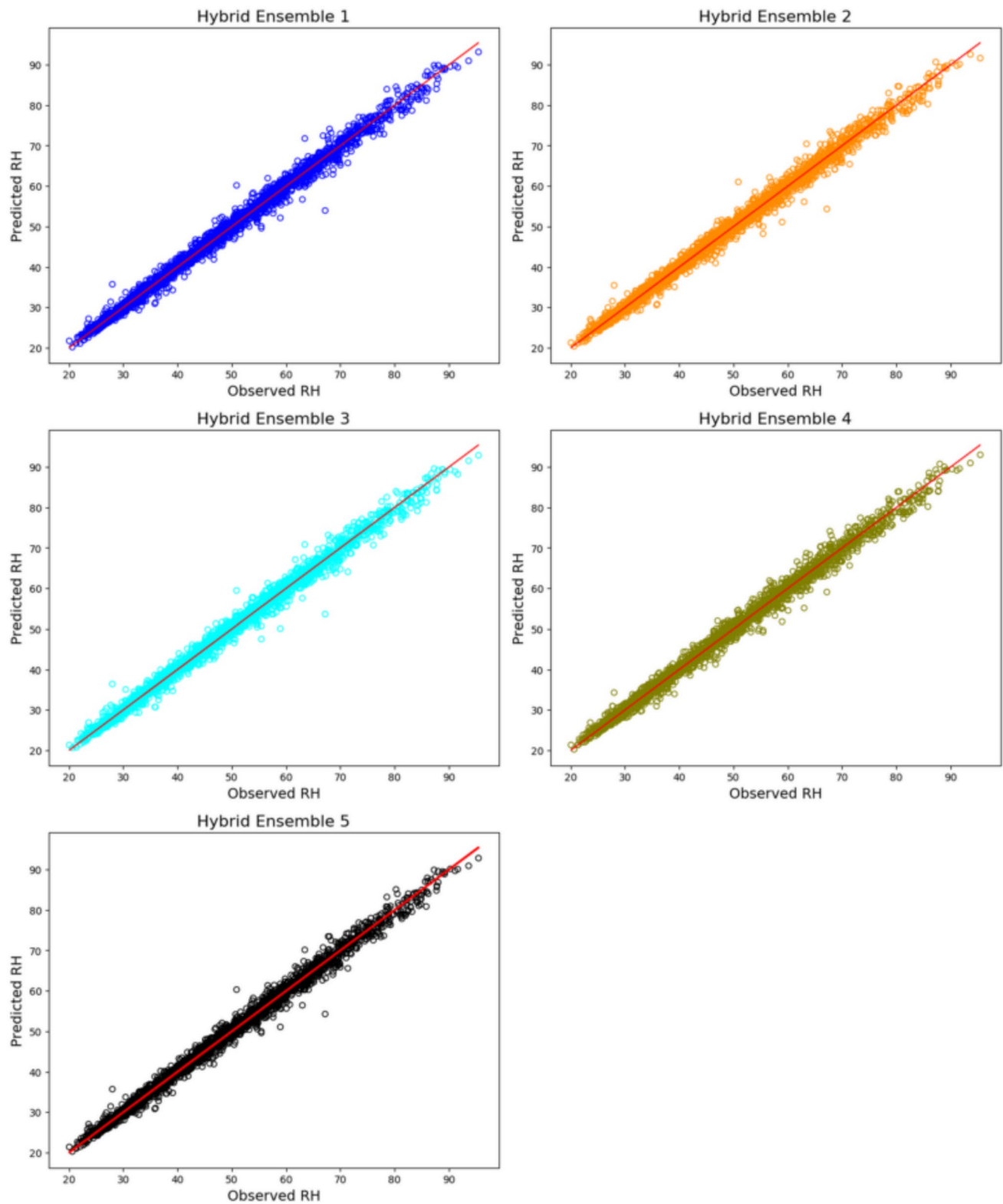


Fig. 4 Scatter plots of daily predicted vs observed relative humidity in Kuwait using HELMs



hinders the development of robust ensemble models, as they often lack the diversity needed to capture the full spectrum of climatic variability (Georgakakos et al. 2005; Trentini et al. 2020). Temporal and spatial variability also impacts the accuracy of humidity predictions. High variability can lead to substantial discrepancies between modeled and actual observations, particularly in regions susceptible to rapid climatic changes. Attempting to forecast at fine temporal scales presents increased uncertainty and a heightened risk of model bias (Maher et al. 2019). Moreover, models initialized from current climate states still face hurdles relating to accuracy in the presence of chaotic climate phenomena (Kushnir et al. 2019). The nonlinear relationships intrinsic to climate systems pose additional challenges. Machine learning procedures must account for complex interactions within climatic variables to avoid oversimplifications that could lead to underestimation of uncertainty. For instance, multivariate models need to delineate significant interactions among temperature, humidity, and precipitation, which often exhibit nonlinear responses to external forcings (Ding et al. 2018).

Recent literature highlights issues with existing ensemble forecasts, noting that larger ensemble sizes can improve prediction skill but are computationally intensive and may not guarantee model reliability due to inherent biases in climate models (Monerie et al. 2023; Milinski et al. 2020). Innovative methodologies, such as utilizing analog climate conditions derived from historical data, may help increase the predictive power of current models (Ding et al. 2018). The developed models hold particular relevance in the context of Kuwait's changing climate patterns. Accurate humidity predictions are increasingly crucial for public health management, urban planning adaptations, workplace safety regulations, and emergency response systems (Alahmad et al. 2023; Alkandari 2024). The high accuracy achieved by these models provides a reliable foundation for decision-making in these critical areas, potentially improving the effectiveness of climate adaptation strategies in arid regions. In addition to long-term urban planning, the high precision of these models makes them suitable for integration into real-time early warning systems, smart city infrastructure, and mobile health advisory platforms. Such integration can aid local governments and public health agencies in issuing timely alerts during high-risk humidity and heatwave periods.

While the models demonstrate excellent performance, several areas warrant further investigation. The current monthly aggregation, while effective, could be enhanced to provide daily predictions, and the modeling framework could be extended to cover broader geographical areas. Incorporating additional environmental parameters and real-time data streams presents opportunities for future research, particularly in developing more comprehensive climate prediction systems for arid regions (Najafi et al.

2021; Chen et al. 2023b). Continued development of these modeling approaches could significantly enhance the ability to predict and respond to changing climate conditions in arid environments.

Conclusion

Given the critical need for accurate humidity forecasting in arid environments, this study conducted a comprehensive analysis of four state-of-the-art EL models to determine their effectiveness in predicting daily humidity levels in Kuwait City. This study collected over a 40-year period of daily climatic data for 11 variables, including Tavg, Tmax, Tmin, SI, WS10M, WS50M, WBT, DF, PS, PR, and RH, and analyzed multicollinearity among these factors. Variables such as Tavg and PS, which exhibited high multicollinearity, were excluded from the training process to improve model stability and accuracy. Performance of the models were assessed using standard statistical performance metrics, including R^2 , RMSE, MAE, and MAPE. Results indicated that all four EL models achieved high predictive accuracy, with R^2 values exceeding 0.99. Among them, the RF demonstrated the highest accuracy, achieving an R^2 of 0.994796 along with the lowest error metrics, which was followed by LightGBM and XGBoost. GBR, however, showed the least effectiveness in prediction. To further improve predictive accuracy, this study constructed five HELMs by combining different sets of RF, XGBoost, GBR, and LightGBM. Among these, HELM-05 (RF + GBR + XGBoost + LightGBM) exhibited the most precise performance, with the highest R^2 (0.99364) and the lowest error metrics, confirmed by the alignment of observed and predicted humidity values in scatter plot analyses. Although HELM-01 (RF + XGBoost + GBR) was the least accurate of the five models, it still showed better performance over standalone EL models.

Overall, this study underscores the potential of HELMs in improving the accuracy of climatic predictions in arid regions, demonstrating the importance of addressing multicollinearity and integrating multiple modeling approaches. The findings provide a solid foundation for future research in this area, suggesting that the integration of ensemble learning with deep learning models could further refine the precision of climatic forecasts. This research not only contributes to the field of climate modeling but also offers actionable insights for improving climate resilience and adaptation strategies in arid regions facing increasing temperature extremes due to climate change.

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Data availability Data will be available on reasonable request.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

Ethical approval This article does not contain any studies with human participants or animals performed by any of the authors.

Consent to participate Not applicable.

Consent for publication Not applicable.

Declaration of generative AI and AI-assisted technologies in the writing process In preparing this work, the authors employed ChatGPT, Grammarly, Claude, and Microsoft Word's online language editing tools to refine the language and punctuation. After utilizing these tools, the authors carefully reviewed and edited the content as needed. The authors assume full responsibility for the final content of the publication.

References

- Acharya N, Shrivastava NA, Panigrahi BK, Mohanty UC (2014) Development of an artificial neural network based multi-model ensemble to estimate the northeast monsoon rainfall over south peninsular India: an application of extreme learning machine. *Clim Dyn* 43:1303–1310
- Ahmed K, Sachindra DA, Shahid S, Iqbal Z, Nawaz N, Khan N (2020) Multi-model ensemble predictions of precipitation and temperature using machine learning algorithms. *Atmos Res* 236:104806
- Alahmad B, Al-Hemoud A, Al-Bouwarthan M, Khraishah H, Kamel M, Akrouf Q, Koutrakis P (2023) Extreme heat and work injuries in Kuwait's hot summers. *Occup Environ Med* 80(6):347–352
- Alahmad B, Shakarchi AF, Khraishah H, Alseaidan M, Gasana J, Al-Hemoud A, Fox MA (2020) Extreme temperatures and mortality in Kuwait: who is vulnerable? *Sci Total Environ* 732:139289
- Al-Dalahmah AK, Al-Hurban AE (2019) The impact of urbanization expansion on the geomorphology of the southern coastal sabkhas from Ras Al-Jailiaha to Al-Khiran, South Kuwait. *J Geogr Inf Syst* 11(05):609
- Al-Dousari AE, Mishra A, Singh S (2023) Land use land cover change detection and urban sprawl prediction for Kuwait metropolitan region, using multi-layer perceptron neural networks (MLPNN). *Egypt J Remote Sens Space Sci* 26(2):381–392
- Alkandari AJ (2024) Climate and climate change aspects of Kuwait. Terrestrial environment and ecosystems of Kuwait: assessment and restoration. Springer Nature Switzerland, Cham, pp 57–91
- Alonso Á, Torres A, Dorronsoro JR (2015) Random forests and gradient boosting for wind energy prediction. In: Hybrid artificial intelligent systems: 10th international conference, HAIS 2015, Bilbao, Spain, June 22–24, 2015, Proceedings 10. Springer International Publishing, pp 26–37
- Alshboul O, Shehadeh A, Almasabha G, Almuflih AS (2022) Extreme gradient boosting-based machine learning approach for green building cost prediction. *Sustainability* 14(11):6651
- Alwadi Y, Alahmad B (2024) The burden of heat in arid regions of the Middle East: an analysis from Jordan and Kuwait. *Environ Res Health* 2(3):035006
- Bentéjac C, Csörgő A, Martínez-Muñoz G (2021) A comparative analysis of gradient boosting algorithms. *Artif Intell Rev* 54:1937–1967
- Boateng EY, Otoo J, Abaye DA (2020) Basic tenets of classification algorithms K-nearest-neighbor, support vector machine, random forest and neural network: a review. *J Data Anal Inf Process* 8(4):341–357
- Carmona P, Climent F, Momparler A (2019) Predicting failure in the US banking sector: an extreme gradient boosting approach. *Int Rev Econ Finance* 61:304–323
- Chen L, Han B, Wang X, Zhao J, Yang W, Yang Z (2023a) Machine learning methods in weather and climate applications: a survey. *Appl Sci* 13(21):12019
- Chen L, Zhong X, Zhang F, Cheng Y, Xu Y, Qi Y, Li H (2023b) FuXi: a cascade machine learning forecasting system for 15-day global weather forecast. *NPJ Clim Atmos Sci* 6(1):190
- Chen T (2015) Xgboost: extreme gradient boosting. R package version 0.4-2, 1(4)
- Coffel ED, Horton RM, De Sherbinin A (2017) Temperature and humidity based projections of a rapid rise in global heat stress exposure during the 21st century. *Environ Res Lett* 13(1):014001
- Cramer MN, Gagnon D, Laitano O, Crandall CG (2022) Human temperature regulation under heat stress in health, disease, and injury. *Physiol Rev* 102(4):1907–1989
- Crawford J, Venkataraman K, Booth J (2019) Developing climate model ensembles: a comparative case study. *J Hydrol* 568:160–173
- Davis RE, McGregor GR, Enfield KB (2016) Humidity: a review and primer on atmospheric moisture and human health. *Environ Res* 144:106–116
- Dietterich T (2000) Ensemble methods in machine learning, pp 1–15
- Ding H, Newman M, Alexander M, Wittenberg A (2018) Skillful climate forecasts of the tropical Indo-Pacific Ocean using model-analogs. *J Climate* 31(14):5437–5459
- Ding Y, Li Z, Zhang C, Ma J (2019) Prediction of ambient pm2.5 concentrations using a correlation filtered spatial-temporal long short-term memory model. *Appl Sci* 10(1):14
- Donat M, Mahmood R, Cos J, Ortega P, Doblas-Reyes F (2024) Improving the forecast quality of near-term climate projections by constraining internal variability based on decadal predictions and observations. *Environ Res Clim* 3(3):035013
- Dong X, Yu Z, Shi Y, Ma Q (2020) A survey on ensemble learning. *Nature Comput Sci* 10(2):123–140
- Dong Y, Ding H (2025) Bayesian optimization of hybrid quantum LSTM in a mixed model for precipitation forecasting. *Mach Learn Sci Technol* 6(1):015065
- Eom H, Son Y, Choi S (2020) Feature-selective ensemble learning-based long-term regional pv generation forecasting. *IEEE Access* 8:54620–54630



- Fan GF, Zhang LZ, Yu M, Hong WC, Dong SQ (2022) Applications of random forest in multivariable response surface for short-term load forecasting. *Int J Electr Power Energy Syst* 139:108073
- Farhat N (2018) Effect of relative humidity on evaporation rates in Nabatieh region. *Leban Sci J* 19(1):59–66
- Faurès J, Goodrich D, Woolhiser D, Sorooshian S (1995) Impact of small-scale spatial rainfall variability on runoff modeling. *J Hydrol* 173(1–4):309–326
- Georgakakos K, Bae D, Jeong C (2005) Utility of ten-day climate model ensemble simulations for water resources applications in Korean watersheds. *Water Resour Manage* 19(6):849–872
- Glaser J, Lemery J, Rajagopalan B, Diaz HF, García-Trabanino R, Taduri G, Johnson RJ (2016) Climate change and the emergent epidemic of CKD from heat stress in rural communities: the case for heat stress nephropathy. *Clin J Am Soc Nephrol* 11(8):1472–1483
- Gleckler P, Taylor K, Doutriaux C (2008) Performance metrics for climate models. *J Geophys Res Atmos*. <https://doi.org/10.1029/2007JD008972>
- González-Sopeña JM, Pakrashi V, Ghosh B (2021) An overview of performance evaluation metrics for short-term statistical wind power forecasting. *Renew Sustain Energy Rev* 138:110515
- Grönquist P, Yao C, Ben-Nun T, Dryden N, Dueben P, Li S, Hoefler T (2021) Deep learning for post-processing ensemble weather forecasts. *Philos Trans R Soc Math Phys Eng Sci* 379(2194):20200092
- Gunawardhana LN, Al-Rawas GA, Kazama S (2017a) An alternative method for predicting relative humidity for climate change studies. *Meteorol Appl* 24(4):551–559
- Gunawardhana L, Al-Rawas G, Kazama S (2017b) An alternative method for predicting relative humidity for climate change studies. *Meteorol Appl* 24(4):551–559
- Guo CY, Chang KH (2022) A novel algorithm to estimate the significance level of a feature interaction using the extreme gradient boosting machine. *Int J Environ Res Public Health* 19(4):2338
- Hasan M, Hasan M, Rahman P (2024) Comparison of rnn-lstm, tfidf and stacking model approach for weather forecasting in Bangladesh using historical data from 1963 to 2022. *PLoS ONE* 19(9):e0310446
- He Z, Brajard J, Wang Y, Wang X, Shen Z (2023) Enhancing climate predictions with combination of dynamical model and artificial neural network
- Hussain A, Tripura S, Aslam A (2024) Rainfall and daily average temperature prediction using machine learning: a case study of Bangladesh
- Javadinejad S, Dara R, Jafary F (2020) Potential impact of climate change on temperature and humidity related human health effects during extreme condition. *Saf Extreme Environ* 2:189–195
- Johnson F, Sharma A (2009) Measurement of gcm skill in predicting variables relevant for hydroclimatological assessments. *J Clim* 22(16):4373–4382
- Jolly W (2007) Sensitivity of a surface fire spread model and associated fire behaviour fuel models to changes in live fuel moisture. *Int J Wildl Fire* 16(4):503
- Jose DM, Vincent AM, Dwarakish GS (2022) Improving multiple model ensemble predictions of daily precipitation and temperature through machine learning techniques. *Sci Rep* 12(1):4678
- Kamal HJ, Thomas M, Abdulhadi A (2024) Perspectives of geography, environment, and physiography of Kuwait. In: *Terrestrial environment and ecosystems of kuwait: assessment and restoration*, Springer Nature Switzerland, Cham, pp 21–55
- Kandasamy O, N M, E S, Kannan B (2025) Rainfall prediction using artificial neural networks and machine learning algorithms over the Coimbatore region. *J Water Clim Change* 16(3):946–959
- Kilibarda M, Hengl T, Heuvelink G, Gräler B, Pebesma E, Tadić M, Bajat B (2014) Spatio-temporal interpolation of daily temperatures for global land areas at 1 km resolution. *J Geophys Res Atmos* 119(5):2294–2313
- Kushnir Y, Scaife A, Arritt R, Balsamo G, Boer G, Doblas-Reyes F, Wu B (2019) Towards operational predictions of the near-term climate. *Nature Clim Change* 9(2):94–101
- Ladi T, Jabalameli S, Sharifi A (2022) Applications of machine learning and deep learning methods for climate change mitigation and adaptation. *Environ Plann B Urban Anal City Sci* 49(4):1314–1330
- Lam R, Sanchez-Gonzalez A, Willson M, Wirsberger P, Fortunato M, Alet F, Battaglia P (2023) Learning skillful medium-range global weather forecasting. *Science* 382(6677):1416–1421
- Lee J, Wang W, Harrou F, Sun Y (2020) Reliable solar irradiance prediction using ensemble learning-based models: a comparative study. *Energy Convers Manage* 208:112582
- Liang M, Chang T, An B, Duan X, Du L, Wang X, Miao J, Xu L, Gao X, Zhang L, Li J, Gao H (2021) A stacking ensemble learning framework for genomic prediction. *Front Genet*. <https://doi.org/10.3389/fgene.2021.600040>
- Lin L, Liang Y, Liu L, Zhang Y, Xie D, Yin F, Ashraf T (2022) Estimating PM_{2.5} concentrations using the machine learning RF-XGBoost model in guanzhong urban agglomeration, China. *Remote Sens* 14(20):5239
- Lingampally V, Radhika K (2023) A cascading ensemble with custom subset generation and multi-level fusion for enhanced breast cancer detection. *Int J Multidiscip Res* 5(3)
- Liu J, Yan J (2024) Research on maximum temperature prediction based on ARIMA–LSTM–XGBoost weighted combination model. *J Circuits Syst Comput* 33(07):2450123
- Ma X, Fang C, Ji J (2020) Prediction of outdoor air temperature and humidity using xgboost. *Iop Conf Ser Earth Environ Sci* 427(1):012013
- Maher N, Milinski S, Suárez-Gutiérrez L, Botzet M, Dobrynin M, Kornblueh L, Marotzke J (2019) The max planck institute grand ensemble: enabling the exploration of climate system variability. *J Adv Model Earth Syst* 11(7):2050–2069
- Mantilla JAO, Cárdenas EJE (2024) Prediction of PM_{2.5} and PM₁₀ concentrations using XGBoost and LightGBM algorithms: a case study in Lima, Peru. *Interfases* 020:183–206
- Masood A, Hameed MM, Srivastava A, Pham QB, Ahmad K, Razali SFM, Baowidan SA (2023) Improving PM_{2.5} prediction in New Delhi using a hybrid extreme learning machine coupled with snake optimization algorithm. *Sci Rep* 13(1):1–17. <https://doi.org/10.1038/s41598-023-47492-z>
- Mathur S, Kumar S, Choudhury T (2024) Prognostication of weather patterns using meteorological data and ml techniques. *Eai Endorsed Trans Energy Web* 11
- Mendes-Moreira J, Soares C, Jorge A, Sousa J (2012) Ensemble approaches for regression. *ACM Comput Surv* 45(1):1–40
- Mienye I, Sun Y (2022) A survey of ensemble learning: concepts, algorithms, applications, and prospects. *IEEE Access* 10:99129–99149
- Miles J (2005) Tolerance and variance inflation factor. In: Everitt BS, Howell DC (eds) *Encyclopedia of statistics in behavioral science*. <https://doi.org/10.1002/0470013192.bsa683>
- Milinski S, Maher N, Olonscheck D (2020) How large does a large ensemble need to be? *Earth Syst Dyn* 11(4):885–901
- Monerie P, Robson J, Ndiaye C, Song C, Turner A (2023) Cmp6 skill at predicting interannual to multi-decadal summer monsoon precipitation variability. *Environ Res Lett* 18(9):094002
- Najafi H, Robertson AW, Bavani ARM, Irannejad P, Wanders N, Wood EF (2021) Improved multi-model ensemble forecasts of Iran's precipitation and temperature using a hybrid



- dynamical-statistical approach during fall and winter seasons. *Int J Climatol* 41(12):5698–5725
- Ncongwane KP, Botai JO, Sivakumar V, Botai CM (2021) A literature review of the impacts of heat stress on human health across Africa. *Sustainability* 13(9):5312
- Nie P, Roccotelli M, Fanti MP, Ming Z, Li Z (2021) Prediction of home energy consumption based on gradient boosting regression tree. *Energy Rep* 7:1246–1255
- Nourani V, Uzelaltinbulat S, Sadikoglu F, Behfar N (2019) Artificial intelligence based ensemble modeling for multi-station prediction of precipitation. *Atmosphere* 10(2):80
- Obisesan O (2024) Machine learning models for prediction of meteorological variables for weather forecasting. *Int J Environ Clim Change* 14(1):234–252
- Ozbek A, Sekertekin A, Bilgili M, Arslan N (2021) Prediction of 10-min, hourly, and daily atmospheric air temperature: comparison of LSTM, ANFIS-FCM, and ARMA. *Arab J Geosci* 14:1–16
- Perne M, Kocijan J, Božnar M, Grašič B, Mlakar P (2023) Hybrid forecasting of wind for air pollution dispersion over complex terrain. *J Environ Inf*
- Putra M, Utami E (2024) Comparative analysis of hybrid model performance using stacking and blending techniques for student drop out prediction in mooc. *Jurnal Resti (Rekayasa Sistem Dan Teknologi Informasi)* 8(3):346–354
- Rader J, Barnes E (2023) Optimizing seasonal-to-decadal analog forecasts with a learned spatially-weighted mask. *Geophys Res Lett.* <https://doi.org/10.1029/2023GL104983>
- Radhika Y, Shashi M (2009) Atmospheric temperature prediction using support vector machines. *Int J Comput Theor Eng* 1(1):55
- Rasp S, Lerch S (2018) Neural networks for postprocessing ensemble weather forecasts. *Mon Weather Rev* 146(11):3885–3900
- Ren Q, Li N, Zhang W (2022) Research on sand-dust storm forecasting based on deep neural network with stacking ensemble learning. *IEEE Access* 10:111855–111863
- Ruiz-Álvarez M, Gomariz-Castillo F, Alonso-Sarria F (2021) Evapotranspiration response to climate change in semi-arid areas: Using random forest as multi-model ensemble method. *Water* 13(2):222
- Sagi O, Rokach L (2018) Ensemble learning: a survey. *Wires Data Min Knowl Discov.* <https://doi.org/10.1002/widm.1249>
- Salam R, Islam ARMT (2020) Potential of RT, bagging and RS ensemble learning algorithms for reference evapotranspiration prediction using climatic data-limited humid region in Bangladesh. *J Hydrol* 590:125241
- Sampath V, Shalakhti O, Veidis E, Efobi JAI, Shamji MH, Agache I, Nadeau KC (2023) Acute and chronic impacts of heat stress on planetary health. *Allergy* 78(8):2109–2120
- Singh N, Chaturvedi S, Akhter S (2019) Weather forecasting using machine learning algorithm. In: 2019 international conference on signal processing and communication (ICSC), IEEE, pp 171–174
- Slingo J, Palmer T (2011) Uncertainty in weather and climate prediction. *Philos Trans R Soc Lond A Math Phys Eng Sci* 369(1956):4751–4767
- Smarra F, Jain A, De Rubeis T, Ambrosini D, D’Innocenzo A, Mangharam R (2018) Data-driven model predictive control using random forests for building energy optimization and climate control. *Appl Energy* 226:1252–1272
- Sobolewski A, Młynarczyk M, Konarska M, Bugajska J (2021) The influence of air humidity on human heat stress in a hot environment. *Int J Occup Saf Ergon* 27(1):226–236
- Sulaiman MS, Abood MM, Sinnakaudan SK, Shukor MR, You GQ, Chung XZ (2021) Assessing and solving multicollinearity in sediment transport prediction models using principal component analysis. *ISH J Hydraul Eng* 27(sup1):343–353
- Taha AA, Malebary SJ (2020) An intelligent approach to credit card fraud detection using an optimized light gradient boosting machine. *IEEE Access* 8:25579–25587
- Tahsin MS, Al Karim M, Ahmed MU, Rahman Y, Tafannum F, Abdullah S (2021) Comparative analysis of weather prediction using ensemble learning models and neural network. In: 2021 19th OITS international conference on information technology (OCIT), IEEE, pp 325–330
- Trentini F, Aalbers E, Fischer E, Ludwig R (2020) Comparing inter-annual variability in three regional single-model initial-condition large ensembles (smiles) over europe. *Earth Syst Dyn* 11(4):1013–1031
- Wang J, Liu H, Zheng G, Li Y, Yin S (2023) Short-term load forecasting based on outlier correction, decomposition, and ensemble reinforcement learning. *Energies* 16(11):4401
- White-Newsome JL, Sánchez BN, Jolliet O, Zhang Z, Parker EA, Dvonch JT, O’neill MS (2012) Climate change and health: indoor heat exposure in vulnerable populations. *Environ Res* 112:20–27
- Wolkoff P (2018) Indoor air humidity, air quality, and health—an overview. *Int J Hyg Environ Health* 221(3):376–390
- Wolkoff P, Azuma K, Carrer P (2021) Health, work performance, and risk of infection in office-like environments: the role of indoor temperature, air humidity, and ventilation. *Int J Hyg Environ Health* 233:113709
- Wu H, Zhou H, Long M, Wang J (2023) Interpretable weather forecasting for worldwide stations with a unified deep model. *Nat Mach Intell* 5(6):602–611
- Wu Z, Yin H, He H, Li Y (2021) Hybrid (dynamical-statistical) models to improve seasonal drought prediction: application to regional meteorological droughts in China
- Xu L, Yu H, Chen Z, Du W, Chen N, Huang M (2023) Hybrid deep learning and s2s model for improved sub-seasonal surface and root-zone soil moisture forecasting. *Remote Sens* 15(13):3410
- Yousaf M, Ali Z, Mohsin M, Ilyas M, Shakeel M (2023) Development of a new hybrid ensemble method for accurate characterization of future drought using multiple global climate models. *Stoch Environ Res Risk Assess* 37(12):4567–4587
- Zhang H, Loáiciga H, Sauter T (2024) A novel fusion-based methodology for drought forecasting. *Remote Sens* 16(5):828
- Zhang L, Xu L, Jiang M, He P (2023) A novel hybrid ensemble model for hourly PM_{2.5} concentration forecasting. *Int J Environ Sci Technol* 20(1):219–230. <https://doi.org/10.1007/s13762-022-03940-3>
- Zhao W, Li J, Zhao J, Zhao D, Zhu X (2019) PDD_GBR: research on evaporation duct height prediction based on gradient boosting regression algorithm. *Radio Sci* 54(11):949–962
- Zoest V, Osei F, Hoek G, Stein A (2019) Spatio-temporal regression kriging for modelling urban no₂ concentrations. *Int J Geogr Inf Sci* 34(5):851–865

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